

Vinh Ly

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Summary

Staff engineer with **18 years** building large-scale distributed systems across Google, YouTube, Whatnot, and Microsoft. Deep hands-on experience with **distributed systems, API design, cloud infrastructure, and data privacy** at companies serving hundreds of millions to billions of users. Spent the last several years leading engineering teams and shipping features that directly shaped product outcomes, which put me in regular conversation with product managers, cross-functional partners, and stakeholder groups around scope, trade-offs, and customer impact. Looking to bring that combination of technical depth and communication to a solutions engineering role.

Core Skills

Languages: C++, Java, Python, Go, TypeScript, SQL

Distributed Systems: gRPC, Protocol Buffers, Thrift, REST APIs, Apache Kafka, ZeroMQ, distributed caching

Infrastructure: Kubernetes, Docker, GCP (BigQuery, Spanner, Pub/Sub, GKE), AWS (EC2, Lambda, S3, RDS)

Observability: Prometheus, Grafana, OpenTelemetry, Dapper tracing, custom metrics pipelines

Security / Privacy: OAuth 2.0, JWT, PII handling, GDPR/CCPA compliance review, PCI-DSS adjacent payment flows

Data: Relational databases (Postgres, MySQL, Spanner), data pipelines, large-scale data processing

Work Experience

Staff Software Engineer | YouTube | Remote | 02/2025 - Present

- Leading the video recommendations infrastructure serving over 2 billion users, owning the systems that surface and rank content across watch pages and the home feed.
- Designed and shipped A/B experiments evaluating new ML ranking features. Results showed measurable lift in watch time and engagement across a segment of the global user base.
- Collaborated daily with product managers and ML researchers to scope ranking changes, assess trade-offs between model complexity and serving latency, and translate experimental outcomes into product decisions.

Staff Software Engineer | Whatnot | Remote | 05/2024 - 02/2025

- Took over ownership of the payments and fraud prevention services serving a live marketplace with millions of users and a growing volume of real-money transactions.
- Rebuilt the payment processing pipeline in Go to improve transaction reliability and reduce the fraud detection gap that had caused chargeback issues in the previous architecture.
- Introduced structured logging and observability tooling across the payments stack, which cut time-to-root-cause during payment incidents from over an hour to under 15 minutes.

Staff Software Engineer / Manager | Google | Remote | 03/2015 - 04/2024

- Served as technical lead for a 40-engineer infrastructure team owning a core piece of Google's internal compute platform, handling resource scheduling and capacity allocation for a significant fraction of Google's workloads.
- Led the migration of an aging internal service framework to a modern, containerized architecture. The migration brought **compute costs down roughly 30%** while simplifying the deployment model for application teams across the company.

- Rewrote the capacity management service in C++ and Python, replacing a brittle hand-rolled solution with one built on top of established internal frameworks. Oncall pages dropped from roughly 40 per quarter to under 10 after the rollout.
- Mentored a group of engineers on C++ correctness, systems design, and incident response. Several were later promoted into senior and staff roles under this guidance.
- Worked with SREs and security teams on regular compliance reviews and access-control audits for the internal platform's control plane.

Software Engineer | Google | Remote | 02/2011 - 02/2015

- Built and maintained backend services in C++ for a Google product used by hundreds of millions of people, focusing on the data ingestion and serving path.
- Introduced a structured testing framework for a subsystem that had been relying on ad-hoc checks. Test coverage went from under 50% to around 80%, and the subsystem became one of the more stable parts of the service in subsequent quarters.
- Worked with a small team to reduce p99 latency on a hot-path query from roughly 120 ms to under 40 ms by rewriting the cache lookup logic and adding a prefetch layer.

Software Design Engineer II | Microsoft | Hybrid | 10/2007 - 12/2011

- Developed and shipped features for a developer-facing product used by internal and external engineers, iterating on the feature set based on user feedback and usage data.
- Reduced build times for a key internal workflow from 20 minutes to under 4 minutes by introducing an incremental compilation strategy and better parallelization across the build pipeline.

Software Design Engineer Intern | Microsoft | Hybrid | 06/2006 - 08/2006

- Intern project: designed and implemented a testing tool used by the team to validate API responses across a set of integration test scenarios.

Education

BS Computer Science | Stanford University | 2003 - 2007